

Question

Partial electrode potentials are shown below:

Half-reaction	Standard Electrode Potential
$4\text{AO}_2(\text{aq}) + 4\text{H}^+ + 6\text{e}^- \rightarrow \text{A}_4\text{O}_6^{2-} + 2\text{H}_2\text{O}$	0.51V
$2\text{AO}_2(\text{aq}) + 2\text{H}^+ + 4\text{e}^- \rightarrow \text{A}_2\text{O}_3^{2-} + \text{H}_2\text{O}$	0.40V
$*2\text{AO}_3^{2-} + 3\text{H}_2\text{O} + 4\text{e}^- \rightarrow \text{A}_2\text{O}_3^{2-} + 6\text{OH}^-$	-0.58V
$*\text{AO}_4^{2-} + 4\text{H}_2\text{O} + 6\text{e}^- \rightarrow \text{A} + 8\text{OH}^-$	-0.66V
$\text{B}_2 + 2\text{e}^- \rightarrow 2\text{B}^-$	0.535V

Note: The standard electrode potentials of reactions marked with "*" are values at pH=14; in this problem, the temperature is assumed to be 298K, and the two-stage ionization constants of H_2AO_3 are 1.3×10^{-2} and 6.3×10^{-8} , respectively.

Then, the equilibrium constant K_1 for the titration of B_2 by $\text{A}_2\text{O}_3^{2-}$ (one of the products is $\text{A}_4\text{O}_6^{2-}$) and the equilibrium constant K_2 for the reaction $\text{AO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{AO}_3$ are respectively (retain two significant figures, an error within 5% is considered correct):

A. All other options are incorrect

B. 5.3×10^{13} , 1.7

C. 5.3×10^{13} , 1.5

D. 5.3×10^{14} , 1.7

E. 5.3×10^{14} , 1.5

F. 5.3×10^{15} , 1.7

G. 5.3×10^{15} , 1.5

H. 3.5×10^{13} , 1.7

I. 3.5×10^{13} , 1.5

J. 3.5×10^{14} , 1.7

K. 3.5×10^{14} , 1.5

L. 3.5×10^{15} , 1.7

M. 3.5×10^{15} , 1.5

N. 5.3×10^{13} , 1.9

O. 5.3×10^{14} , 1.9

P. 5.3×10^{15} , 1.9

Q. 3.5×10^{13} , 1.9

R. 3.5×10^{14} , 1.9

S. 3.5×10^{15} , 1.9

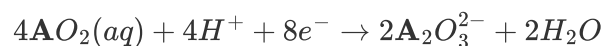
Answer

Correct Answer: **F**

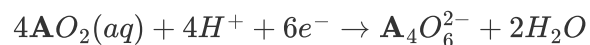
Detailed Explanation

Calculate K_1

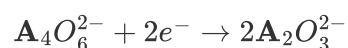
Connect the battery in series with a standard hydrogen electrode, and calculate the change in Gibbs free energy of the reaction (hydrogen electrode changes are omitted here)



$$\Delta_r G_m^\circ = -nEF = -308.752 kJ/mol$$



$$\Delta_r G_m^\circ = -nEF = -295.244 kJ/mol$$



$$\Delta_r G_m^\circ = -13.508 kJ/mol$$

Therefore $E = 0.070V$

CHECKPOINT

2 PTS

The standard electrode potential of $\mathbf{A}_4O_6^{2-} + 2e^- \rightarrow 2\mathbf{A}_2O_3^{2-}$ is 0.070V

Therefore, the electromotive force of $\mathbf{A}_2O_3^{2-}$ titrating $\mathbf{B}_2$ is 0.465V,

$$K_1 = e^{\frac{nEF}{RT}} = 5.3 \times 10^{15}$$

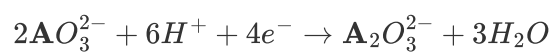
CHECKPOINT

1 PTS

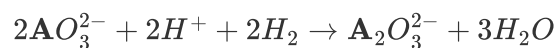
The equilibrium constant K_1 is 5.3×10^{15}

Calculate K_2

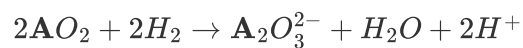
Convert the data under alkaline conditions:



$$E' = 0.6623\text{V}$$

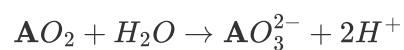


$$K = 6.051 \times 10^{44}$$



$$K = 1.114 \times 10^{27}$$

Subtract to get



$$K = 1.357 \times 10^{-9}$$

CHECKPOINT

1 PTS

The equilibrium constant of $\text{AO}_2 + \text{H}_2\text{O} \rightarrow \text{AO}_3^{2-} + 2\text{H}^+$ is $K = 1.357 \times 10^{-9}$

Thus we obtain



$$K = 1.656$$

CHECKPOINT

2 PTS

The equilibrium constant of $\text{A}\text{O}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{A}\text{O}_3$ is $K_2 = 1.656$